

# B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

## ME 19-205-0404 APPLIED THERMODYNAMICS (2019 Scheme)

(Use of steam table is permitted)

(Data not given may be suitably assumed. All such assumptions may be clearly stated.)

Maximum Marks: 60

Time: 3 Hours

### Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Learn first and second law of thermodynamics and the application of these laws for various thermodynamic processes.
- CO2: Apply the knowledge of thermodynamic laws to various engineering systems.
- CO3: Analyse the performance of steam generators with additional accessories used for enhancing the performance.
- CO4: Understand the various types of nozzles and its performance at various back pressure conditions and also for different flow conditions.
- CO5: Calculate the mole and mass fractions of various mixtures and also various specific properties of the mixtures.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

### PART A

(Answer ALL questions)

(8 × 3 = 24)

|                                                                                                                                                       | Marks | BL | CO | PO |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|----|----|
| I. (a) Write steady flow energy equation for a single stream entering and single stream leaving a control volume and explain the various terms in it. | 3     | L2 | 1  | 1  |
| (b) Establish the Inequality of Clausius.                                                                                                             | 3     | L1 | 2  | 1  |
| (c) Explain the concept of quality or dryness fraction of steam.                                                                                      | 3     | L1 | 2  | 1  |
| (d) Draw phase equilibrium diagram on p-v coordinates for a substance which shrinks in volume on melting.                                             | 3     | L1 | 3  | 1  |
| (e) Explain binary vapour cycles with a neat sketch.                                                                                                  | 3     | L2 | 3  | 1  |
| (f) Explain velocity compounding in an impulse turbine with the help of a neat sketch.                                                                | 3     | L1 | 4  | 1  |
| (g) Describe various laws for mixtures of ideal gases with respect to partial pressure, partial volume, enthalpy, entropy and internal energy.        | 3     | L2 | 5  | 1  |
| (h) Differentiate between ideal, perfect and real gases.                                                                                              | 3     | L2 | 5  | 1  |

### PART B

(4 × 12 = 48)

- II. (a) Show that the efficiency of a reversible engine is independent of the nature or amount of working substance.
- (b) A reversible heat engine delivers 65 kW power and rejects energy at the rate of 40 kJ/s to a reservoir at 27°C. Draw a schematic sketch and determine the efficiency of the engine and the temperature at which energy is absorbed by the engine.

OR

(P.T.O.)

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|           |                                                                                                                                                                                                                                                                                                                                                                                             | Marks | BL | CO | PO    |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|----|-------|
|           |                                                                                                                                                                                                                                                                                                                                                                                             |       | L3 | 2  | 1     |
| III.      | (a) What is meant by 'Exergy'? Write down the exergy equation for a non-flow process.                                                                                                                                                                                                                                                                                                       | 6     | L5 | 1  | 1,2,4 |
|           | (b) A perfect gas is compressed according to the law $p v^{1.25} = \text{constant}$ from an initial pressure of 1 bar and volume of $0.9 \text{ m}^3$ to a final volume of $0.6 \text{ m}^3$ . Determine<br>(i) work done per kg of gas during the process<br>(ii) the change in entropy per kg of gas during the process.<br>Take $\gamma = 1.4$ and $R = 287 \text{ J/kgK}$ .             | 6     |    |    |       |
| IV.       | (a) Why do we try to modify the ideal Rankine cycle as follows. Explain with suitable sketches.<br>(i) By re-heating<br>(ii) By regeneration<br>Mention the practical difficulties associated if any.                                                                                                                                                                                       | 6     | L4 | 2  | 1     |
|           | (b) In a thermal power plant working on Rankine cycle, superheated steam at 10 bar and $350^\circ\text{C}$ is produced in the boiler and the condenser is operated at 0.2 bar. Calculate the quality of the steam at the turbine outlet and the thermal efficiency of the cycle, using Mollier diagram. What is the change in the quantity of steam if the turbine efficiency is 90%.       | 6     | L5 | 3  | 1,2,4 |
| <b>OR</b> |                                                                                                                                                                                                                                                                                                                                                                                             |       |    |    |       |
| V.        | (a) Differentiate between fire tube and water tube boilers with neat sketches.                                                                                                                                                                                                                                                                                                              | 6     | L2 | 3  | 1     |
|           | (b) A boiler evaporates 3.6 kg of water per kg of coal into dry saturated steam at 10 bar. The temperature of feed water is $32^\circ\text{C}$ . Find the equivalent evaporation from and at $100^\circ\text{C}$ . Also find the factor of evaporation.                                                                                                                                     | 6     | L4 | 2  | 1,2   |
| VI.       | (a) Dry saturated steam at 10 bar is expanded through a convergent divergent nozzle. The pressure of steam leaving the nozzle is 0.1 bar. Find:<br>(i) velocity of steam at throat for maximum discharge<br>(ii) the area at exit<br>(iii) steam discharge if the throat area is $1.2 \text{ cm}^2$ .<br>Assume the flow is isentropic and there are no friction losses. Take $n = 1.135$ . | 6     | L5 | 4  | 1,2   |
|           | (b) Explain metastable flow in nozzles with the help of a neat sketch.                                                                                                                                                                                                                                                                                                                      | 6     | L2 | 4  | 1,2   |
| <b>OR</b> |                                                                                                                                                                                                                                                                                                                                                                                             |       |    |    |       |
| VII.      | (a) In a steam turbine, steam enters the convergent divergent (CD) nozzle at 20 bar and $400^\circ\text{C}$ . The discharge pressure of the nozzle is 0.1 bar. The loss due to friction in the divergent section of the nozzle is equivalent to 10% of the total enthalpy drop. Calculate the throat and exit area for a discharge of 14 kg/sec of steam.                                   | 6     | L5 | 4  | 1,2,4 |
|           | (b) What is meant by Degree of reaction (R). Show that for a Parson's reaction turbine $R = \frac{1}{2}$ .                                                                                                                                                                                                                                                                                  | 6     | L2 | 4  | 1,2   |

(Continued)

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|           |                                                                                                                                                                                                                                                                                                                                  | Marks | BL | CO | PO    |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|----|-------|
| VIII.     | (a) Derive an expression for the mean value of gas constant for a mixture of gases.                                                                                                                                                                                                                                              | 6     | L3 | 5  | 1,2   |
|           | (b) A gas mixture consists of 20% O <sub>2</sub> , 30% N <sub>2</sub> and 50% CO <sub>2</sub> on volume basis. Determine the gravimetric analysis of the mixture. Find the equivalent molecular mass and the apparent gas constant of the mixture.                                                                               | 6     | L4 | 5  | 1,2,4 |
| <b>OR</b> |                                                                                                                                                                                                                                                                                                                                  |       |    |    |       |
| IX.       | (a) What is the apparent molar mass for a gas mixture? Does the mass of every molecule in the mixture equal to the apparent molar mass?                                                                                                                                                                                          | 6     | L2 | 5  | 1,2   |
|           | (b) A vessel of 1.8 m <sup>3</sup> capacity contains oxygen at 8 bar and 50°C. The vessel is connected to another vessel of 3.6 m <sup>3</sup> capacity containing carbon monoxide at 1 bar and 20°C. A connecting valve is opened and the gases mix adiabatically. Calculate the final temperature and pressure of the mixture. | 6     | L4 | 5  | 1,2,4 |

Bloom's Taxonomy Levels

L1 – 10%, L2 – 30%, L3 – 15%, L4 – 20%, L5 – 25%.

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